MIL-E-1/1376A(NAVY)
8 September 1970
SUPERSEDING
MIL-E-1/1376(NAVY)
16 June 1961

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING, TYPES 7551, 7558

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of MIL-E-1.

DESCRIPTION: Beam power, rf amplifier, oscillator, or frequency multiplier, F1 = 175 MHz
Outline ---- 6-3 (E1A)
Base ---- E9-1
Envelope ---- T-6-1/2
Cathode ---- Coated unipotential

Base Connections:

Pin Number:	1	2	3	4	5	6	7	8	9
Element:	k	g1	g2	h	h	a	g3	g2	k

ABSOLUTE - MAXIMUM RATINGS:

Parameter: / Units:	Ef V	Eb Vdc	Ec1 Vdc	Ec2 Vdc	Ec3 Vdc	Ehk Vdc	Rg1 Meg
RF Power Amp. and Osc. -Class C Telegraphy and RF Power Amp. -Class C FM Telephony Continuous	Note 9	300	-125	250	0	100	0.1
RF Power Amp. and Osc. -Class C Telegraphy and RF Power Amp. -Class C FM Telephony Intermittent	Note 9	300	-125	250	0	100	0.1
Plate Modulated RF Power Amp. Class C Telephony Continuous	Note 9	250	-125	250	0	100	0.1
Plate Modulated RF Power Amp. Class C Telephony Intermittent	Note 9	250	-125	250	0	100	0.1
Frequency Multiplier Continuous	Note 9	300	-125	250	0	100	0.1
Frequency Multiplier Intermittent	Note 9	300	-125	250	0	100	0.1

TEST CONDITIONS:

7551	13.5	250	-18	250	0	---	---
7558	6.3	250	-18	250	0	---	---

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ABSOLUTE - MAXIMUM RATINGS:

Parameter: Units:	Ib ma	Ic2 ma	Ic1 ma	Pg2 W	Pp W	Pi W	T °C	Alt ft
RF Power Amp. and Osc. -Class C Telegraphy and RF Power Amp. -Class C FM Telephony Continuous	70	15	5	2	10	21	210	(See note 12)
RF Power Amp. and Osc. -Class C Telegraphy and RF Power Amp. -Class C FM Telephony Intermittent	80	15	5	2	12	24	210	(See note 12)
Plate Modulated RF Power Amp. Class C Telephony Continuous	60	10	5	1.4	7	15	210	(See note 12)
Plate Modulated RF Power Amp. Class C Telephony Intermittent	70	10	5	1.4	8	17.5	210	(See note 12)
Frequency Multiplier Continuous	50	15	5	2	10	13	210	(See note 12)
Frequency Multiplier Intermittent	60	15	5	2	12	15	210	(See note 12)
TEST CONDITIONS:								
7551	---	---	---	---	---	---	---	---
7558	---	---	---	---	---	---	---	---

GENERAL:

Qualification - Required

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Method	Requirement or Test	Conditions	AQL %	Insp. Level or Code	Symbol	Min.	Limits		Units
							Bogie	Max.	
	Quality conformance inspection, part 1								
1266	Total grid current	Eb = 180 Vdc; Ec1 = 0; Rk = 170 ohms; Rg = 0.1 Meg	0.65	II	Ic	---	---	-2.0	mAdc
1256	Electrode current (1) (anode)	Eb = 75 Vdc; (see note 1)	0.65	II	Ib	85	---	---	mAdc
1256	Electrode current (2) (anode)	Ec1 = -48 Vdc	0.65	II	Ib	---	---	50	mAdc
1256	Electrode current (1) (screen)	Eb = 75 Vdc; (see note 1)	1.5	II	Ic2	---	---	24.0	mAdc
1201	Short and discontinuity detection		0.25	II	---	---	---	---	---
1251	Audio frequency noise and microphonics	Ec1 = -15 Vdc; Rg1 = 0.5 Meg; Rb = 2000 ohms; Ck = 1.000 uf min; Ecal= 140 mVac; (see note 2)	2.5	I	VU	---	---	17	---
	Quality conformance inspection, part 2								
1211	Insulation of electrodes		4.0	S3	R	100	---	---	Meg
1301	Heater current (7551)		6.5	S3	If	330	360	390	mA
1301	Heater current (7558)		6.5	S3	If	745	800	855	mA
1336	Heater-cathode leakage		6.5	S3	Ihk	---	---	20	uAdc
---	Cathode activation test	Prr = 3 Hz; tp = 25 milliseconds; egk = +10V; Ec1 = -50 Vdc; (see note 3)	6.5	S3	Pik	315	---	---	mA
1256	Electrode current (3) (anode)		6.5	S3	Ib	30	40	50	mAdc
1256	Electrode current (2) (screen)		6.5	S3	Ic2	---	---	7.5	mAdc
1306	Transconductance		6.5	S3	Sm	4200	5300	6400	umhos
---	Power output (1)	Eb = 300 Vdc; Ec2=200 Vdc; Ec1 = 0; Rg1 = .018 Meg; Ib=80 mA max. (see note 4)	6.5	S3	Po	6.5	---	---	W
---	Power output (2)	Ei=Note 10; Eb=300 Vdc; Ec2 = 200 Vdc; Ec1 = 0, Rg1 = .018 Meg; Ib=80 mA max., (see note 4)	6.5	S3	Po	6.0	---	---	W
1331	Capacitance	Without external shield	6.5	S3	Cg1-p	---	---	0.16	pF

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Method	Requirement or Test	Conditions	AQL %	Insp. Level or Code	Symbol	Min.	Limits		Units
							Bogie	Max.	
	Quality conformance inspection, part 2 (Continued)								
1002	Low-pressure voltage breakdown	Pressure = 55 ±5 mm Hg Voltage - 500 Vac	6.5	Note 8	---	---	---	---	---
1031	Low frequency vibration (2)	Rb = 2000 ohms	6.5	S3	Ep	---	---	100	mVac
1041	Shock test	Hammer angle = 30° Ebk = +100 Vdc (see note 5)	20	S2	---	---	---	---	---
1031	Vibration fatigue test	G = 2.5; fixed frequency; F = 25 min.; 60 max.	6.5	See note 8	---	---	---	---	---
	Post shock and fatigue test end points								
---	Heater-cathode leakage		---	---	Ihk	---	---	40	uAdc
---	Total grid current		---	---	Ic1	---	---	40	uAdc
---	Change in transconductance (1) of individual tubes		---	---	Δ_tSm	---	---	15	%
---	Miniature tube glass and base strain	(See note 6)	---	---	---	---	---	---	---
---	Miniature tube base strain	(See note 7)	---	---	---	---	---	---	---
	Quality conformance inspection, part 3								
1501	Intermittent life test	Group B; Ef=Note 11; Ebb = 180 Vdc; Ecc2 = 250 Vdc; Ehk=+180V; RK = 170 ohms; Rgl=0.1 meg. T (envelope) = 210°C. min.	---	---	t:	500	---	---	hrs.
---	Life test end points	Grid current	---	---	Ic1	---	---	-40	uAdc
		Power output (1)	---	---	Po	4.5	---	---	W
	Periodic check test								
1031	Variable frequency vibration (1)	Rp = 2,000	---	---	Ep	---	---	400	mVac

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NOTES:

1. Use square wave knee test set, Model H-1733A-1, or equivalent; 50 percent duty cycle.
2. Calibration: apply 0.14V to plate of tube under test. RB = 2000 ohms. The rejection level shall be set at the VU meter reading obtained during calibration.
3. Maximum allowable slump during the pulse = 50 ma.
4. Test in 175 MHz amplifier circuit; driver power = 1.0 watt.
5. A grid resistor of 0.1 megohm shall be added; however, this resistor shall not be used when a thyratron-type short indicator is employed.
6. Miniature tube glass and base strain test. The miniature tube glass and base strain test shall be performed on a lot sampling basis in accordance with the following double sampling plan:

N1	C1	N2	C1 + C2
48	8	96	20

The sample shall be taken and the test performed not less than 1 week or more than 3 weeks after the lot has been manufactured. This test is a time-dependent test, and the sampling plan is based on the test being made only during the time interval specified herein. The room temperature and boiling water containers for the water baths shall be as shown on figure 1.

The containers shall be at least 3/4 full for each immersion. The holders for this test shall be in accordance with figures 2 through 5, and shall be fitted with deflection cones in accordance with Drawing 200-JAN.

Prior to testing all tubes shall be submitted to pin straightening procedure. Align the axis of the tube with the axis of the deflection cone and carefully push the tube onto the cone until the tube button lies firmly against the cone. Place holder and tubes into a room temperature water bath (25 ±5°C.), so that the tubes attached to the cones are completely submerged in the water for a period of 5 seconds. The holder with the tubes shall then be immediately transferred and properly submerged into a boiling water bath (97° to 100°C.), for a period of 10 seconds. Upon removal from the boiling water bath, the holder with the tubes shall be immediately transferred and properly submerged into a room temperature water bath (25 ±5°C.) for a period of 5 seconds.

Upon removal from this bath, the tubes shall be examined for defects. Evidence of air leaks shall constitute a defective.

A record of the results of this test shall be made available to the Government inspector at the time the lot is submitted for inspection. This record shall be the basis of lot acceptance for this test.

7. Miniature tube base strain test. The miniature tube base strain test shall be performed on a lot sampling basis in accordance with the following double sampling plan:

N1	C1	N2	C1 + C2
35	1	70	4

The sample shall be taken and the test performed not less than 1 week, nor more than 3 weeks after the lot has been manufactured. This test is a time-dependent test, and the sampling plan is based on the test being made only during the time interval specified herein. The deflection cones for this test shall be in accordance with figure 6.

Prior to testing all tubes shall be submitted to pin straightening procedure. Align the axis of the tube with the axis of the deflection cone and carefully push the tube onto the cone until the tube button lies firmly against the cone. Place tubes and cones in an upright position and hold under normal room temperature and humidity conditions for 48 ±4 hours. At the end of this period, the sample shall be examined for defectives. Evidence of air leaks shall constitute a defective. A record of the results of this test shall be made available to the Government inspector at the time the lot is submitted for inspection. This record shall be the basis of lot acceptance for this test.

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8. This test shall be conducted on the initial lot and thereafter on a lot approximately every 6 months.

   Once a lot has passed, the 6 month rule shall apply. In the event of lot failure, the lot shall be rejected and the succeeding lot be subjected to this test. Standard MIL-STD-105, Size Code Letter E, shall apply.

9. The heater voltage is 6.3 ±10 percent Vac for tube type 7558 or 13.5 ±10 percent for tube type 7551.

10. The heater voltage is 5.5 Vac for tube type 7558 or 12.0 Vac for tube type 7551.

11. The heater voltage is 6.3 Vac for tube type 7558 or 15.0 Vac for tube type 7551.

12. See "Reduced pressure (altitude) rating" and altitude, maximum peak voltage.

User activities:
Navy - AS, OS, MC, CG, SH

Preparing activity:
Navy - EC

(Project 5960-N879)

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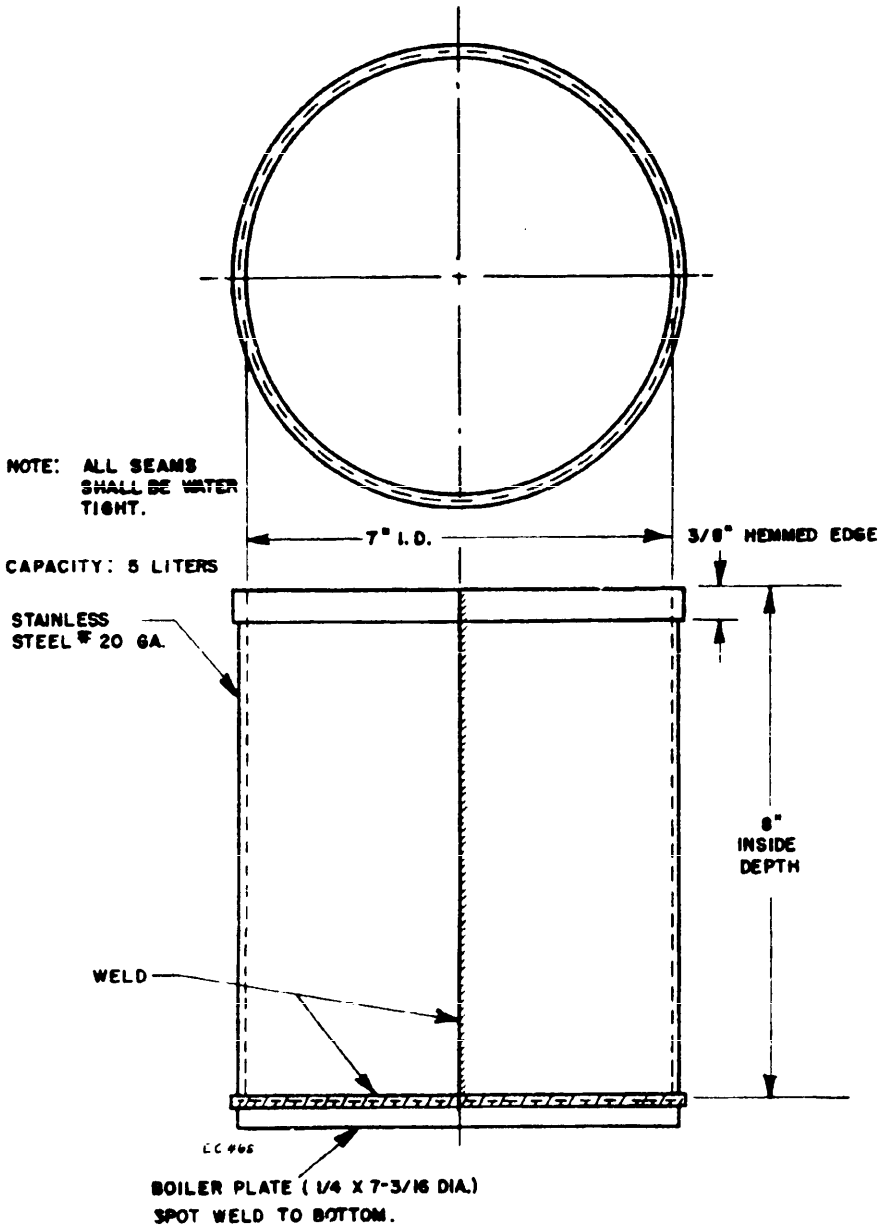


FIGURE 1. Water container.

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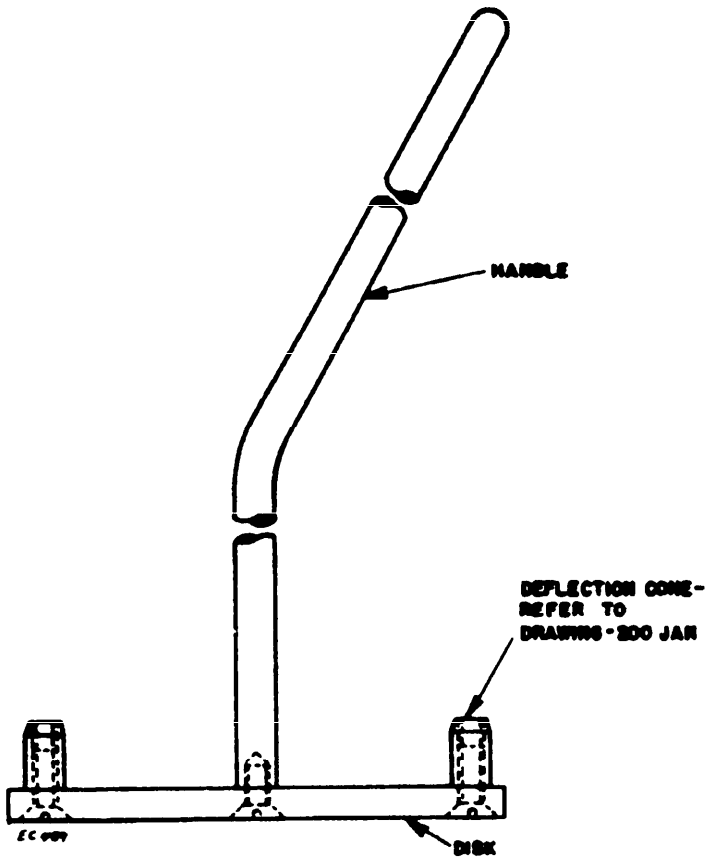


FIGURE 2. Miniature tube glass and base strain test holder assembly.

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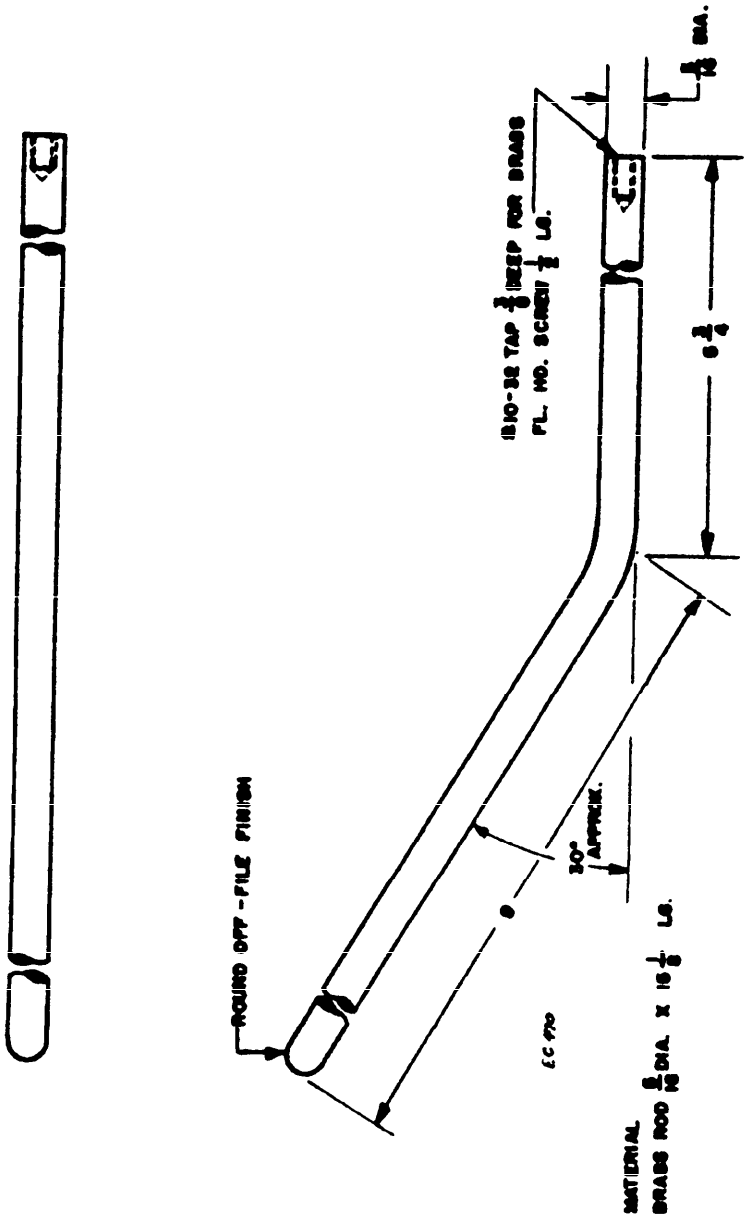


FIGURE 3. Handle.

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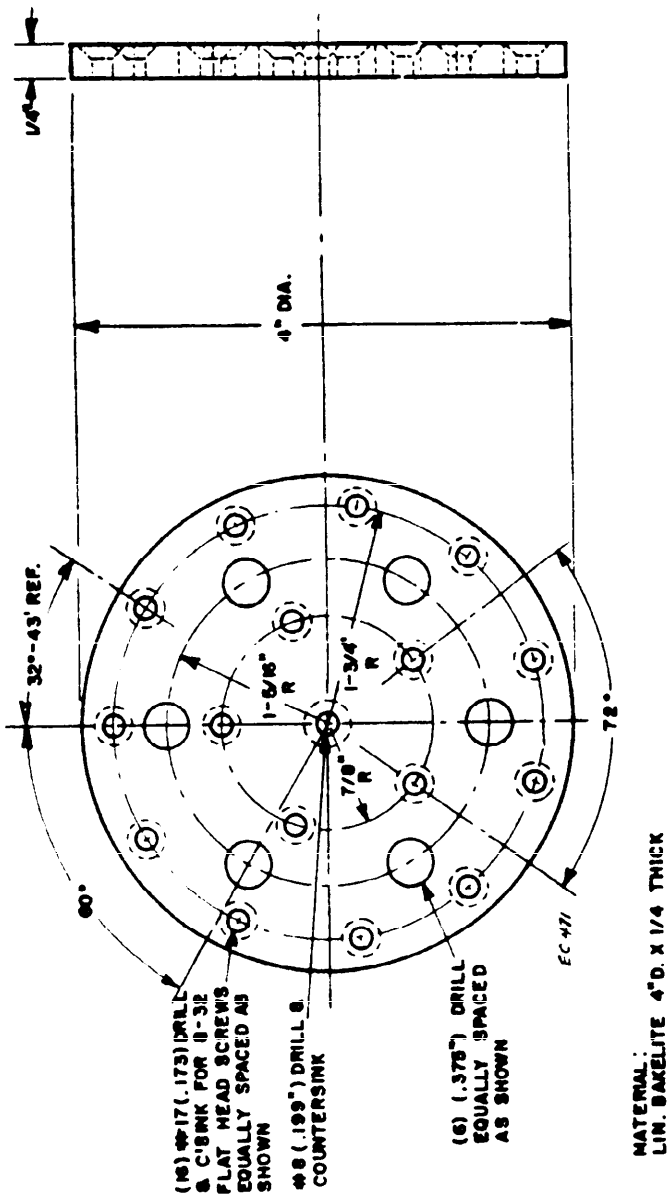


FIGURE 4. Seven pin disk.

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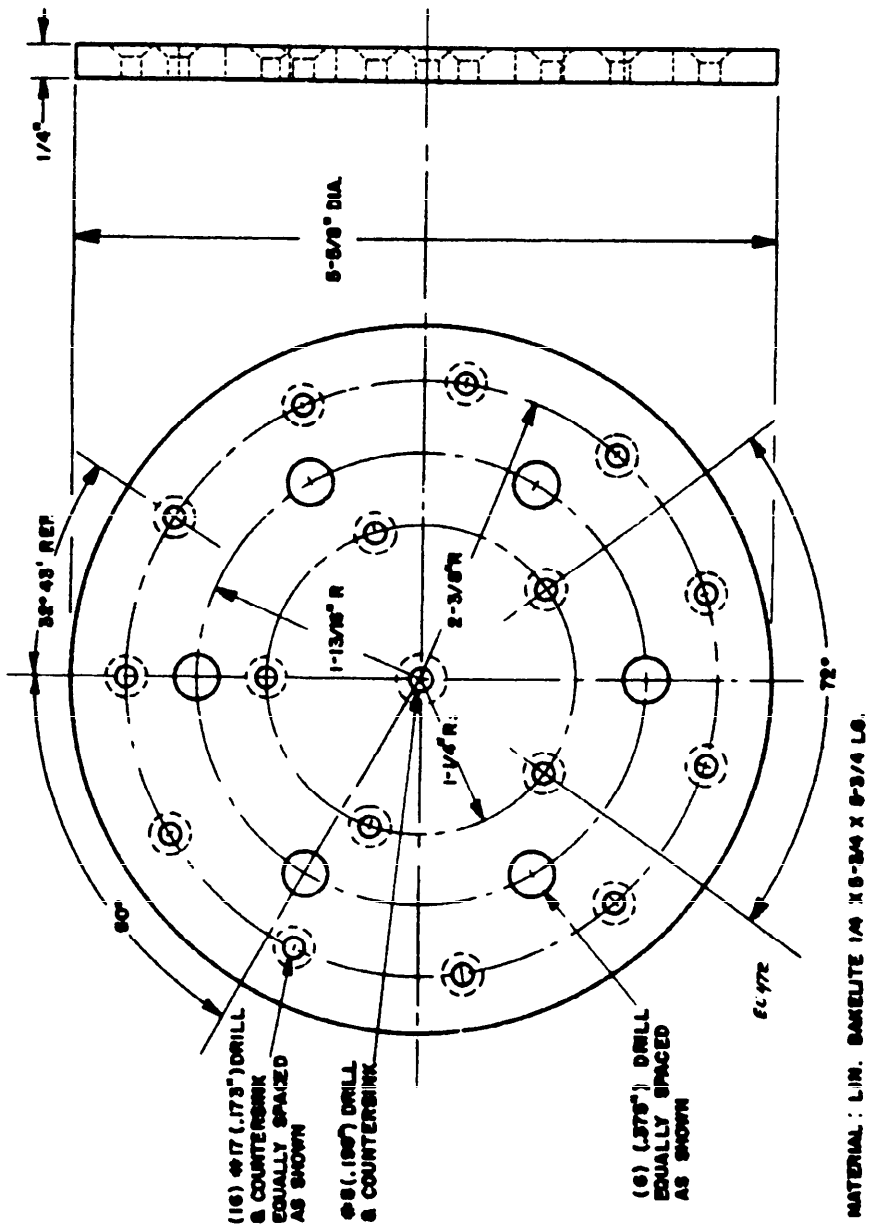


FIGURE 5. Nine pin disk.

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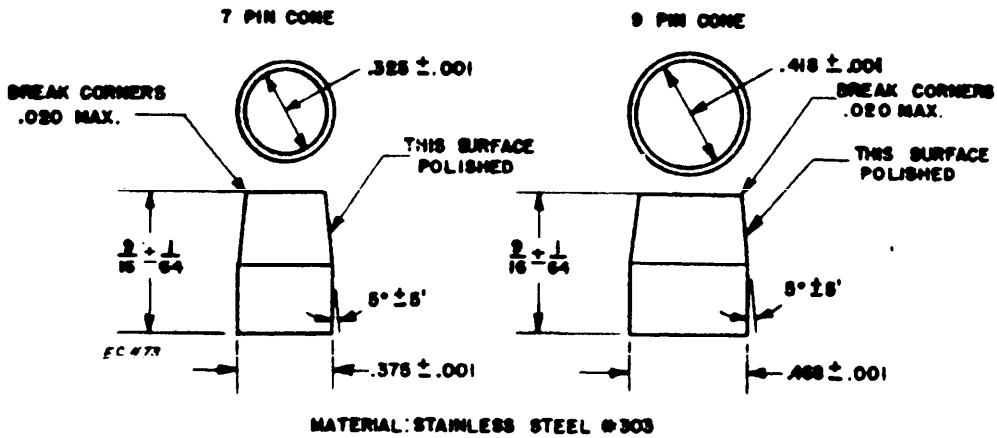


FIGURE 6. Deflection cone for miniature tube base strain test.

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NAVAL ELECTRONIC SYSTEMS COMMAND
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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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